# Aidan Howard

## Design, Build + Research Projects

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# Projects at a Glance

Six projects, ordered as they appear in this portfolio

## 01 DIMLab Hip Exoskeleton

*Univ. of Dayton*

Defined the motor misalignment problem and developed active alignment for a rehab exoskeleton

**Skills:** MATLAB simulation, SolidWorks, research

## 02 Multifunctional Wrench Handle

*UMass Amherst*

Handle for interchangeable wrench heads, checked with hand calcs and FEA

**Skills:** Strength of materials, ANSYS FEA, CAD

## 03 Modular Robotic Fish

*RIT Hydrodynamic Lab*

Waterproof Arduino-controlled fish for testing seal-whisker flow sensors

**Skills:** Iterative prototyping, 3D printing, soldering

## 04 Miniature Piston

*RIT*

Small-scale piston assembly drafted and machined from my own drawings

**Skills:** Mill, lathe, engineering drawings

## 05 Freestanding Climbing Wall

*Personal Project*

Climbing wall designed to stand without bolting into the walls of my home

**Skills:** Power tools, basic structural design

## 06 Automated Chime Machine

*RIT*

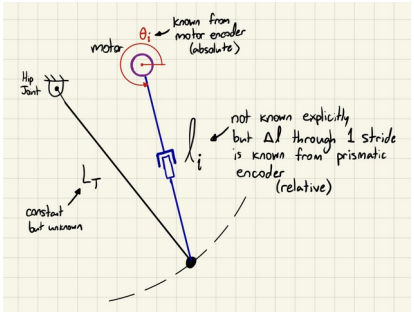
Led a 4-person team to design, fabricate, and program an automated instrument

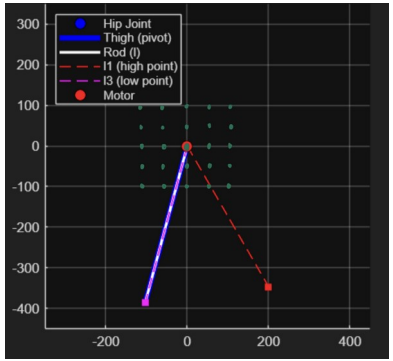
**Skills:** Team leadership, fabrication, mechanisms

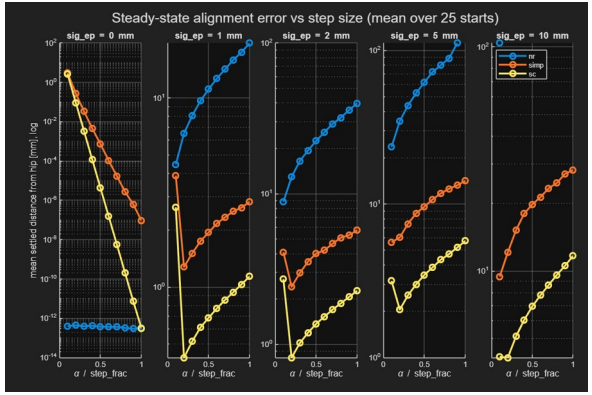
# DIMLab Hip Exoskeleton

Univ. of Dayton, Summer 2026 | Defined the misalignment problem and developed active motor-to-hip-joint alignment

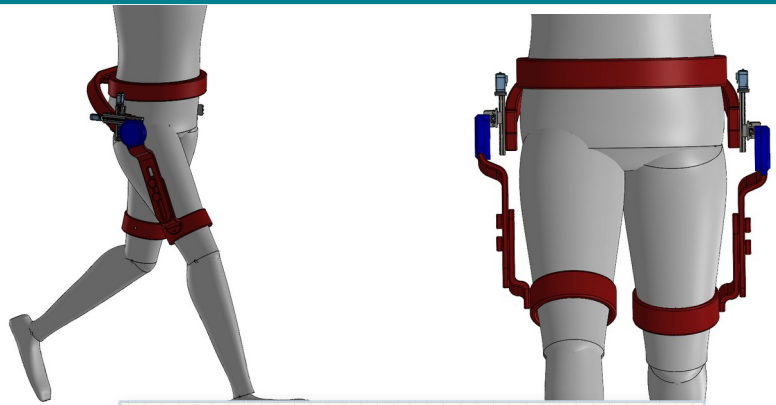
## Algorithms + Data Analysis + MATLAB Simulation

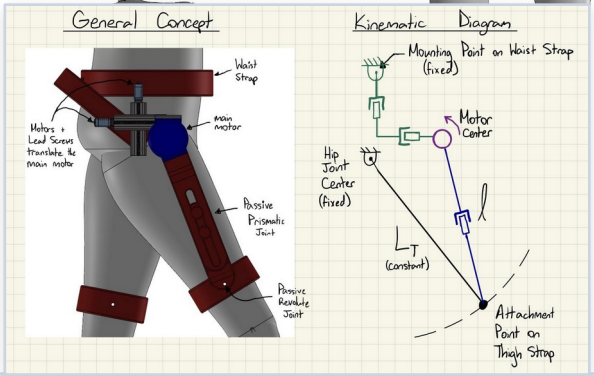



## CAD Modeling

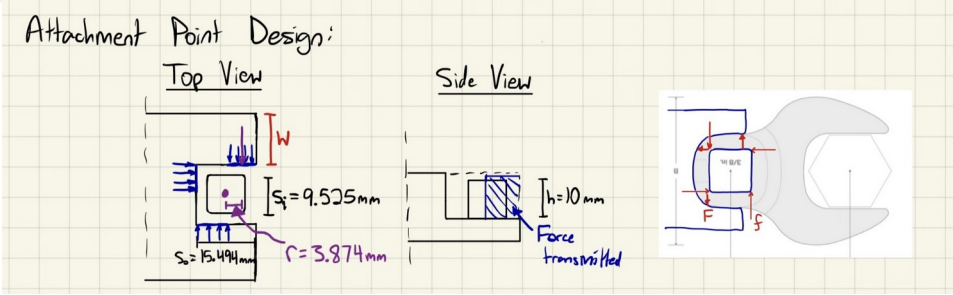




# Multifunctional Wrench Handle

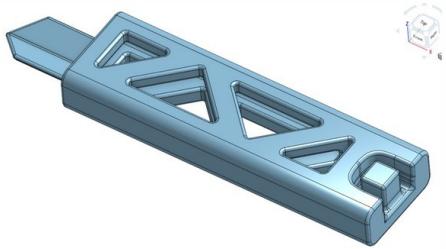
UMass Amherst | Loads sized from published torque-tension data; ANSYS FEA with the wrench head modeled in contact, validated against physical testing

## Attachment Point Design

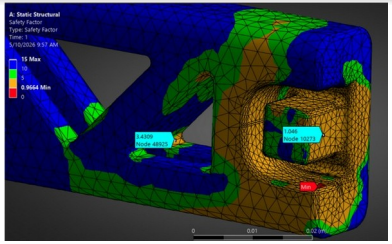


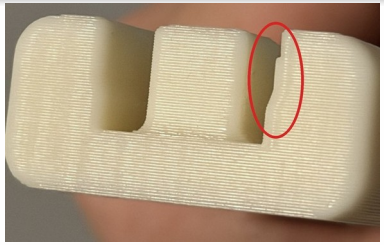

Hand estimation of the forces transmitted from the wrench head into the socket, used to approximate the shear and bending stresses in the outer walls of the attachment point.

## CAD Model



## ANSYS FEA vs. Physical Test



Minimum safety factor of 0.97 at the head pocket matched the slight plastic deformation after testing (circled)

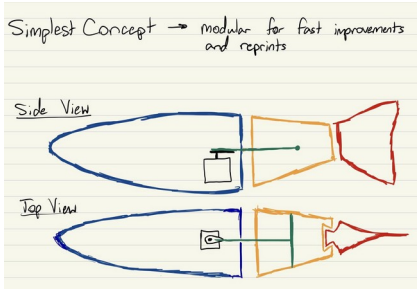
## Physical Test



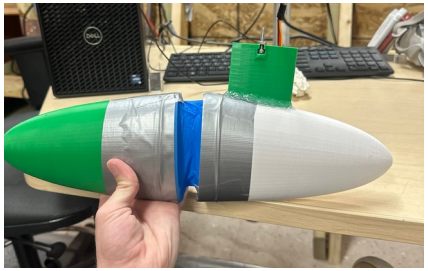
# Modular Robotic Fish

RIT Hydrodynamic Lab | Led design + fabrication of an Arduino-driven fish that simulates wakes for sensor testing

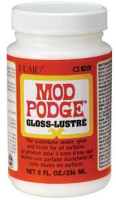
## Concept




## Fabrication

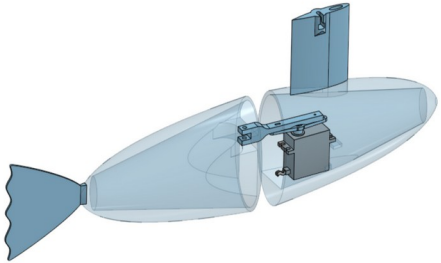


## Revision

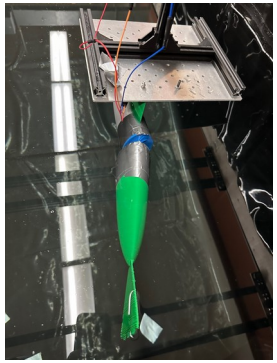



(Water seeped through layers in the plastic during testing; sealed all printed parts with Mod Podge)

## CAD Model



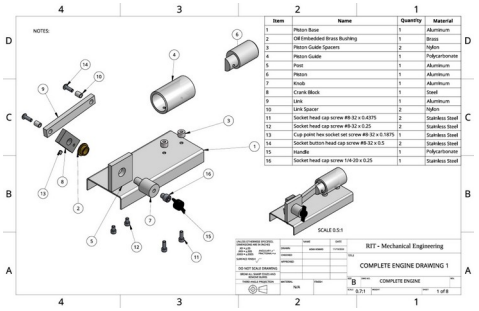
## Testing

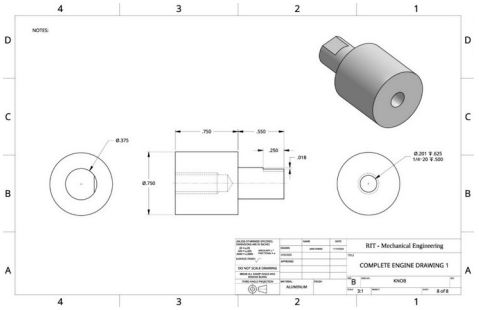


# Hands-On Builds: Miniature Piston + Climbing Wall

RIT: drafted piston drawings, then machined parts on mill + lathe | Personal: freestanding wall built with power tools

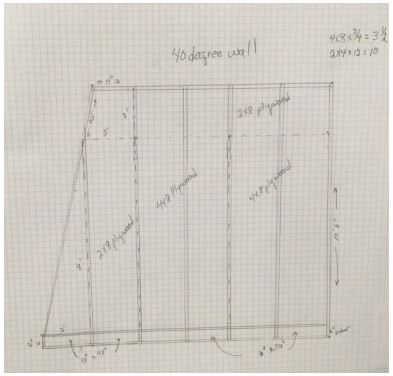
## Miniature Piston Project






## Freestanding Climbing Wall



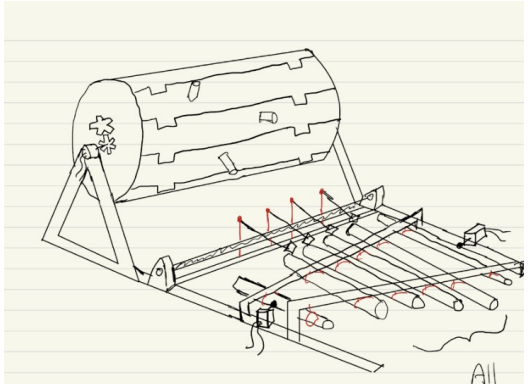




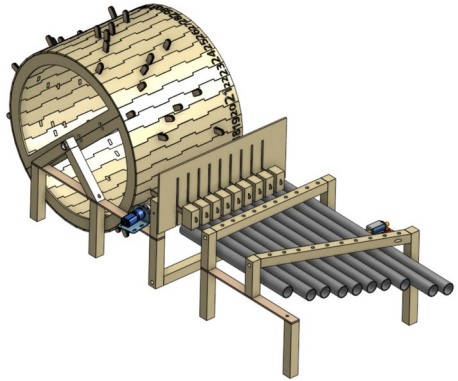

# Automated Chime Machine

RIT | Led a 4-person team to design, fabricate, and program an automated musical instrument

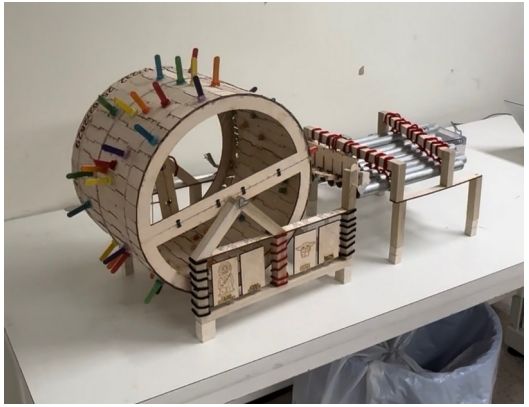
## Concept



## CAD Model



## Fabrication



**Outcome:** Selected by professors and TAs as a showcase project for the ImagineRIT innovation exhibition

# Thank You

## Contact

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